

BIOLOGICAL IMPLICATIONS OF SULFIDE IN SEDIMENT—A REVIEW FOCUSING ON SEDIMENT TOXICITY

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Abstract—The biological implications of sulfide in sediment are poorly understood and all too often ignored despite the fact that sulfide can be extremely important in determining sediment toxicity to resident biota. Sulfide influences sediment toxicity in three major ways, which are reviewed in detail: as a toxicant in its own right; by reducing metal toxicity by forming insoluble metal sulfide solids and/or by forming metal sulfide complexes; and by affecting animal behavior, which in turn can alter the toxicity of not just the sulfide but also other sediment contaminants. Our present limited understanding of sulfide in sediments represents two major problems related to determining the toxicity of sediments, both in the laboratory and the field, and the causative agents of such toxicity. First, we do not know how important sulfide toxicity is to resident populations. Second, by not adequately considering sulfide toxicity, we risk underestimating toxicity and misidentifying the causative agents. Generic and specific recommendations related to resolving these problems are provided, including appropriate measurement and monitoring of sulfide in the laboratory and the field, determination of toxicity thresholds and tolerances for a wide range of sediment-dwelling organisms, further development of toxicity identification evaluation procedures, further research into sulfide effects on metal toxicity, and determination of the influence of sulfide on bioirrigation.

Keywords—Sulfide Sediment Toxicity Metals Adaptation Bioirrigation

INTRODUCTION

Sulfide, produced by anaerobic decomposition of organic matter with sulfate as the electron acceptor, is a common, and often abundant, constituent of aquatic sediments. Benthic invertebrates living on or in sediments are commonly exposed to this substance, which, because it can be extremely toxic [1,2], can also be a pollutant of concern. However, the biological implications of sulfide in sediments are poorly understood and all too often ignored [3] despite the fact that acid-volatile sulfide (AVS) has been intensively studied and incorporated into contaminated-sediment assessment models [4–6].

In contrast, ammonia, another naturally occurring sediment toxicant, is much better understood. For example, the latest U.S. Environmental Protection Agency and U.S. Army Corps of Engineers guidance on testing dredged material [7] addresses this issue in detail and includes an appendix dedicated to ammonia toxicity. But, for sulfide, all that is stated is that “The chemistry and toxicology of sulfides is less well-understood than that of ammonia. However, sulfides are not likely to be a problem in most open-water situations, or in bioassays where adequate oxygen levels are maintained in the overlying water” [7]. These two sentences ignore both the in situ effects of sulfide in sediments and the fact that bioassays with field-collected sediments are all too often designed to obviate sulfide toxicity.

Many cases of sediment toxicity observed during standardized toxicity testing were caused by ammonia in addition to metals or organic chemicals [8,9]. However, sulfide is more abundant and more toxic to aquatic organisms than ammonia (e.g., at 25°C and pH 8, the U.S. Environmental Protection Agency [2] fresh- and saltwater quality criterion for hydrogen

sulfide is 2 µg/L H₂S, whereas that for unionized ammonia is 35 µg/L NH₃). Thus, sulfide may well be more important than ammonia in determining sediment toxicity to resident biota.

The purpose of this article is to review what is known of sulfide in sediments and to suggest what needs to be done to adequately assess its biological significance, particularly in relation to sediment toxicity measurements. We specifically review the three major ways in which sulfide influences sediment toxicity: (1) as a toxicant in its own right; (2) by reducing metal toxicity by forming insoluble metal sulfide solids and/or by forming metal sulfide complexes; and (3) by affecting animal behavior, which in turn can alter the toxicity of not just sulfide but also other sediment contaminants.

TOXICITY OF SULFIDE TO BENTHIC INVERTEBRATES

Sulfide production occurs in sediments containing large amounts of organic matter once oxygen is depleted by aerobic bacteria. Typically this occurs a few millimeters to centimeters beneath the sediment surface, depending on the influx of organic matter. Marine waters contain relatively high sulfate concentrations (28 mM [10]). Thus, in marine sediments, sulfate reduction by sulfate-reducing bacteria (e.g., *Desulfovibrio* and *Desulfotomaculum*) is the dominant process producing large amounts of sulfide via the decomposition of organic matter. Although sulfate concentrations in freshwater are much lower than in marine waters (0.12 mM in typical river water [10]), high levels of sedimentary organic matter, particularly in lakes, are sufficient to produce measurable sulfide. As a result, micromolar to millimolar levels of sulfide have been measured in natural fresh- and marine pore waters. Figure 1 provides typical profiles of sulfate, sulfide, and dissolved oxygen in both overlying and pore waters of aquatic sediments. Not shown in this figure is the fact that, under some circumstances, the sulfide zone may extend into the water column [11,12].

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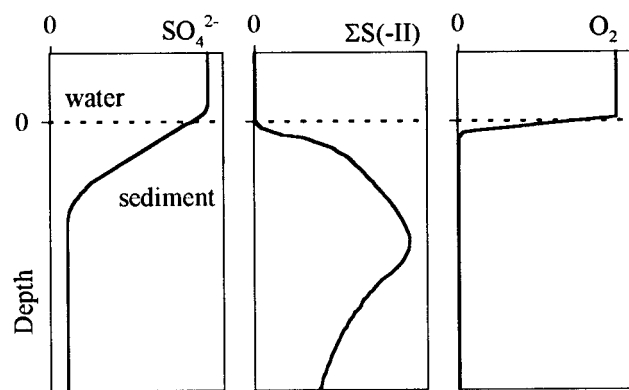


Fig. 1. Typical profiles of sulfate (SO_4^{2-}), total sulfide ($\Sigma\text{S}(-\text{II})$) and oxygen (O_2) in the overlying water and pore water of aquatic sediments.

The toxicity of sulfide is pH dependent, as is the case for ammonia (and some other contaminants). Sulfide exists mainly in two forms in water: unionized hydrogen sulfide (H_2S) and sulfide ion (HS^-) (the third form, S^{2-} , is never a dominant aqueous species [13]):



where $\log K_{a1} = -7.02$ [14]. At pH 8.0 (typical of seawater), about 9% of sulfide is in the form of H_2S , whereas at pH 6.0, approx. 91% of sulfide is in the form of H_2S . At pH 7.0, $\text{H}_2\text{S} = \text{HS}^-$. Neutral molecular H_2S can readily diffuse across the nonaqueous cell membrane into the animal, whereas the HS^- ion may have difficulty in penetrating the strong charged membrane surface. Vismann [15] found that it was exclusively H_2S that diffused into the shrimp *Crangon crangon* and caused toxicity. However, HS^- was found to penetrate (although less readily than H_2S) the membrane and cause additional toxicity to the fathead minnow, *Pimephales promelas* [16], and to the echiuran worm, *Urechis caupo* [17].

Toxicity thresholds for sulfide have been set for fish and other water-column organisms [1,2]. Although studies have been conducted on the effects of sulfide on benthic invertebrates (most of which were conducted in the absence of sediment; Table 1), general toxicity thresholds have not been determined for benthic organisms. These have not been set because it is extremely difficult to obtain a reasonable dose-effect relationship. The difficulties lie in two unique aspects of sulfide. First, because H_2S is volatile and HS^- is readily oxidized [18], it is difficult to maintain a constant sulfide concentration during routine laboratory toxicity testing, which generally lasts several days and involves aeration. Second, the presence of sulfide is often, if not always, accompanied by a lack of oxygen; both can contribute to toxicity [19]. For instance, even with water-only exposure under flow-through conditions, concentrations of sulfide and/or dissolved oxygen are not stable [20,21]. Hence, caution is advised when the effects data listed in Table 1 are used. Nevertheless, these effects data reveal a strong potential for sulfide to cause toxicity in many sediments, because sulfide concentrations in pore waters are frequently higher than effects concentrations [22].

A recently developed computer-operated exposure system by Vismann [23] may provide a reasonable technique to determine sulfide thresholds to benthic organisms. In this water-only exposure system, sulfide, oxygen, and pH are automatically regulated and monitored by specific electrodes. However,

difficulties in maintaining a constant (free) sulfide concentration in the presence of sediment remain.

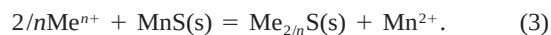
Sulfide toxicity to benthic animals is also significantly affected by their habitats. Animals from sulfide-rich environments generally show high resistance to sulfide because of adaptation [19,21]. Such adaptation can be found at sulfide- and metals-rich deep-sea hydrothermal vents [24]. However, different benthic animals in less extreme environments also show different tolerances to sulfide. Because sulfide is more prevalent in marine sediments than in freshwater sediments, marine benthic invertebrates generally appeared to be more tolerant of sulfide compared with their freshwater counterparts (Table 1). Because of their ability to undergo anaerobiosis (i.e., to maintain basic life support processes using the energy derived through anaerobic metabolism), tubificid oligochaetes appear to be one of the most sulfide-resistant freshwater invertebrates [25], whereas adult bivalves and polychaetes are among the most sulfide-resistant marine invertebrates [19,21].

EFFECTS OF SULFIDE ON METAL TOXICITY IN SEDIMENTS

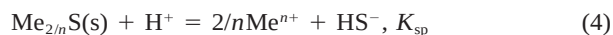
In anoxic sediments, sulfide produced by sulfate reduction reacts with Fe^{2+} and Mn^{2+} to form iron and manganese sulfide solids, including amorphous iron sulfide, mackinawite, greigite, pyrrhotite, troilite, pyrite [26,27], and pink and green manganese sulfides [28]. Except for pyrite, all other iron and manganese monosulfides, which are readily dissolved in acid and hence termed acid-volatile sulfide (AVS), have higher solubility products and can be displaced by other metals to form more insoluble metal sulfides.

Di Toro et al. [4,5] have proposed an AVS/SEM model (SEM indicates simultaneously extracted metals, which are extracted under the same conditions under which AVS content is determined) that recognizes that AVS is a reactive pool of solid-phase sulfide that is available to bind with metals and hence reduce free metal ion concentrations. Their model predicts that certain metals in sediment will not be toxic if the molar concentration of AVS is higher than that of SEM. The AVS/SEM model has been intensively verified for five divalent metals in anoxic sediments (i.e., cadmium, copper, lead, nickel, and zinc [6]) and has recently been tested for a monovalent metal (silver [29]).

The fact that the AVS/SEM model applies only to these few metals does not imply that sulfide has no effect on the toxicity of other metals. In fact, the toxicity of other metals and metalloids that form metal sulfides less soluble than iron and manganese monosulfides in sediments can be reduced by AVS:



If all the metal in sediment is in the form of $\text{Me}_{2/n}\text{S}$ (i.e., AVS is in excess), then the free metal ion activity in pore water is controlled by dissolution of $\text{Me}_{2/n}\text{S}$:



$$\{\text{Me}^{n+}\}^{2/n} = K_{sp, \text{Me}_{2/n}\text{S}} \{\text{Me}_{2/n}\text{S(s)}\} \{\text{H}^+\} / \{\text{HS}^-\} \quad (5)$$

where K_{sp} is the solubility product of $\text{Me}_{2/n}\text{S}$ and $\{i\}$ denotes the activity of species i .

The metal sulfides formed can be either in discrete (pure) $\text{Me}_{2/n}\text{S}$ phases or in a solid solution (i.e., isomorphous replacement of Fe^{2+} by Me^{n+} in the crystalline lattice of FeS) formed

Table 1. Sulfide toxicity to benthic invertebrates

Species	$\Sigma [S(-II)]^a$ (mg/L)	pH	Endpoint ^b	Reference
Freshwater invertebrates				
Amphipod <i>Gammarus</i>	0.002	7.8	65–105-d no mortality	[54]
	0.02	7.8	96-h LC50	[54]
	0.059	7.5	96-h LC50	[55]
Amphipod <i>Crangonyx</i>	0.84	7.4	96-h LC50	[55]
Mayfly <i>Baetis</i>	0.02	7.6	96-h LC50	[55]
Mayfly <i>Hexagenia</i>	<0.015	8.0	138-d no mortality	[56]
	0.06	8.0	12-d LC50	[56]
	0.76	8.0	138-d 100% mortality	[56]
	0.111	7.7	96-h LC50	[55]
	0.165	8.0	96-h LC50	[56]
Mayfly <i>Ephemera</i>	0.316	7.4	96-h LC50	[55]
Isopod <i>Assellus</i>	1.07	7.5	96-h LC50	[55]
Marine invertebrates				
Amphipod <i>Rhepoxynius</i>	1.47	8.0	48-h LOEC	[20]
	1.6	8.0	48-h LC50	[20]
Amphipod <i>Eohaustorius</i>	1.92	8.0	48-h LOEC	[20]
	3.32	8.0	48-h LC50	[20]
Amphipod <i>Anisogammarus</i>	0.2	8.2	96-h LC50	[21]
	3.2	8.2	24-h LC50	[21]
Amphipod <i>Corophium</i>	1.4	8.3	24-h LC50	[21]
Amphipod <i>Gnorimosphaeroma</i>	5.2	8.0	96-h LC50	[21]
Urchin <i>Strongylocentrotus</i>	0.1	8.0	48-h NOEC	[20]
	0.13	8.0	48-h LOEC	[20]
	0.19	8.0	48-h EC50	[20]
Urchin <i>Lytechinus</i>	>0.1	8.0	49-d mortality occurred	[57]
	>10 ^c	8.0	49-d mortality occurred	[57]
Shrimp <i>Crangon</i>	0.64	8.0	1-h LT50	[15]
Crab <i>Cancer</i> , zoeae	0.5	8.1	96-h LC50	[21]
Crab <i>Cancer</i> , first instar	1.0	8.1	96-h LC50	[21]
Mussel <i>Mytilus</i> , embryo	0.05	8.0	48-h NOEC	[20]
	0.09	8.0	48-h LOEC	[20]
	0.1	8.0	48-h EC50	[20]
Mussel <i>Mytilus</i>	1.9	8.0	96-h EC50	[58]
	>50	8.0	96-h LC50	[58]
Clam <i>Macoma</i>	6.0	8.2	96-h LC50	[21]
Clam <i>Arctica</i>	6.4	7.5	10-d LOEC	[59]
Oyster <i>Crassostrea</i>	1.4	8.2	96-h LC50	[21]
Polychaete <i>Nereis</i>	5.76	8.0	24-d LT50	[60]
Polychaete <i>Nereis</i>	4.8	8.0	96-h NOEC	[61]
Polychaete <i>Capitella</i>	>16	8.0	3-h LOEC in settlement time	[62]

^a Concentration expressed as total sulfide ($\Sigma [S(-II)] = [H_2S] + [HS^-]$). At pH 8, $[H_2S] \approx 0.09 \Sigma [S(-II)]$.

^b EC50 = concentration that causes 50% sublethal effect; LC50 = concentration that causes 50% lethal effect (mortality); LOEC = lowest-observed-effect concentration. LT50 = exposure time that causes 50% mortality; NOEC = no-observed-effect concentration.

^c Pore-water total sulfide concentration in the presence of sediment.

by coprecipitation with and/or adsorption on iron monosulfides [28,30,31]. The activity of $Me_{2/n}S(s)$ is identical to 1 when a pure $Me_{2/n}S$ phase forms and far less than 1 when a solid solution forms [28,30]. Equation 5 then becomes

$$\{Me^{n+}\}^{2/n} \leq K_{sp, Me_{2/n}S} \{H^+\}/\{HS^-\} \quad (6)$$

K_{sp} values for a variety of metal sulfides (Table 2) indicate that, besides Cd^{2+} , Cu^{2+} , Pb^{2+} , Ni^{2+} , Zn^{2+} , and Ag^+ , many other metal or metalloid ions, including at least Cu^+ , Co^{2+} , Hg^{2+} , and As^{3+} , can form metal or metalloid sulfides less soluble than iron and manganese monosulfides. The toxicity of these metals in anoxic sulfidic sediments likely is also controlled by sulfide. Although the AVS/SEM model predicts the nontoxicity of certain metals in sediments, the toxic effects of other metals in the presence of AVS have not been well studied.

Although the AVS/SEM model shows predictability of nontoxicity of a few metals in sediments, it does not mean that

the model has fully addressed the role of sulfide in controlling metal toxicity. First, both AVS and SEM in sediment show strong temporal (daily and seasonal) and spatial (horizontal and vertical) variations [6,32–36], and it is particularly difficult to determine where and when to sample and measure AVS and SEM. Second, for some metals that form very strong insoluble metal sulfides, it is not the ratio or difference between AVS and SEM but the presence or absence of AVS that controls free metal ion concentrations in pore water. For example, it has been demonstrated that silver toxicity will not occur as long as there is measurable AVS in sediment [29]. Criticisms of other aspects of the AVS/SEM model can be found elsewhere [34,36–39].

As a “soft-sphere” ligand [10], sulfide in anoxic pore waters also forms strong complexes with “soft” metal ions, including Ag^+ , Cu^+ , Zn^{2+} , Cd^{2+} , Hg^{2+} , and Pb^{2+} and hence reduces the concentrations of the corresponding free metal ions,

Table 2. Solubility products (K_{sp}) of metal/metalloid sulfides (at 1 atmosphere and 25°C; ionic strength = 0)

Metal sulfide	Log K_{sp}	Reference
MnS(s, pink) + $H^+ = Mn^{2+} + HS^-$	2.98	[14]
MnS(s, green) + $H^+ = Mn^{2+} + HS^-$	-0.02	[14]
FeS(s, amorphous) + $H^+ = Fe^{2+} + HS^-$	-2.95	[27]
FeS(s, mackinawite) + $H^+ = Fe^{2+} + HS^-$	-3.6	[27]
FeS(s, greigite) + $H^+ = Fe^{2+} + HS^-$	-4.4	[27]
FeS(s, pyrrhotite) + $H^+ = Fe^{2+} + HS^-$	-5.1	[27]
FeS(s, troilite) + $H^+ = Fe^{2+} + HS^-$	-5.25	[27]
NiS(s, α) + $H^+ = Ni^{2+} + HS^-$	-5.52	[14]
NiS(s, β) + $H^+ = Ni^{2+} + HS^-$	-11.02	[14]
NiS(s, γ) + $H^+ = Ni^{2+} + HS^-$	-12.72	[14]
CoS(s, α) + $H^+ = Co^{2+} + HS^-$	-7.42	[14]
CoS(s, β) + $H^+ = Co^{2+} + HS^-$	-11.72	[14]
ZnS(s, amorphous) + $H^+ = Zn^{2+} + HS^-$	-9.02	[63]
ZnS(s, wurtzite) + $H^+ = Zn^{2+} + HS^-$	-9.68	[63]
ZnS(s, sphalerite) + $H^+ = Zn^{2+} + HS^-$	-11.47	[64]
PbS(s, galena) + $H^+ = Pb^{2+} + HS^-$	-12.25	[65]
CdS(s, greenockite) + $H^+ = Cd^{2+} + HS^-$	-14.36	[66]
$\frac{1}{2} Cu_2S(s, chalcocite) + \frac{1}{2} H^+ = Cu^+ + \frac{1}{2} HS^-$	-17.01	[67]
CuS(s, covellite) + $H^+ = Cu^{2+} + HS^-$	-22.06	[68]
$\frac{1}{2} Ag_2S(s, acanthite) + \frac{1}{2} H^+ = Ag^+ + \frac{1}{2} HS^-$	-18.11	[14]
HgS(s, black) + $H^+ = Hg^{2+} + HS^-$	-38.72	[14]
HgS(s, cinnabar) + $H^+ = Hg^{2+} + HS^-$	-39.22	[14]
$\frac{1}{2} As_2S_3(s, orpiment) = HAsO_3^{2-} + 3/2 HS^- + 7/2 H^+$	-44.45	[69]

which are generally thought to be the most toxic metal species to aquatic biota. However, the thermodynamic formation constants and bioavailability of such metal sulfide complexes are not well known, which has limited our capability to understand the role of sulfide in controlling metal toxicity. The literature is often contradictory. For example, the silver sulfide complex, $AgHS^0$, has been reported as possibly toxic to biota [40]. However, LeBlanc et al. [41] have demonstrated that its toxicity to the fathead minnow, *P. promelas*, is at least 17,500 times less than that of free silver ion, Ag^+ .

EFFECTS OF SULFIDE ON ANIMAL BEHAVIOR

Animals living in sediments can and do adopt a variety of defensive behavioral responses to enable them to avoid prolonged exposure to toxic sulfide. Mobile animals can escape from affected sediment to nearby unaffected sediment and/or to the bottom water, provided that the sulfide concentrations are not immediately toxic and cover only relatively small-scale areas. Benthic macroinvertebrates that dig and live in tubes in sediments can cope with sulfide in sediment pore waters by irrigating these tubes with clean(er) overlying water [42,43].

However, neither escape nor irrigation may be effective when sulfide is present in the bottom water or when the animals are moving within the sediments (i.e., when a new tube has not been established). Benthic animals, particularly those living in sediments contaminated with sulfide, are therefore likely to show some degree of tolerance to sulfide, which may be achieved by (1) selective exclusion of sulfide via anaerobic respiration or possession of H_2S -insensitive enzymes; (2) possession of a cytochrome oxidase and an oxygen-transporting blood pigment, which are insensitive to sulfide; (3) dependence on anaerobic energy metabolism; or (4) detoxification of sulfide [24,44].

Behavioral responses may also have a secondary, but nonetheless important effect, in that they may also alter the toxicity of other contaminants in sediments. Enhanced bioirrigation with oxic overlying water through animals' tubes will create microenvironments that more closely resemble the overlying

water than the surrounding sediments and pore waters. Actively bioirrigating benthic tube dwellers, such as *Hexagenia*, *Sialis*, and *Polycentropus*, have been shown to bioaccumulate cadmium exclusively from overlying lake water, even when the lake sediments they inhabited were artificially heavily contaminated by cadmium [45]. In addition, sulfide stress has been shown to enhance bioirrigation. Injection of sulfide into the tubes of three marine nereid polychaetes caused either an increased duration of ventilation bursts (*Nereis virens* and *N. succinea*) or a decreased duration of rest periods (*N. diversicolor*) [46].

Finally, it is likely that the distributions of benthic invertebrates in sulfide-contaminated sediments are, at least to some extent, governed by their ability to adapt to sulfide toxicity [44,47]. As we learn more about differential sulfide adaptation abilities, we may be able to use the results of in situ biological surveys to determine whether sulfide could be affecting species distributions. For instance, pore-water sulfide constitutes an important factor in the distribution of three species of nereid polychaetes (*N. virens*, *N. diversicolor*, and *N. succinea*) in Danish estuarine sediments [47]. *Nereis virens* was confined to low-sulfide sediments ($<50 \mu M$), *N. succinea* was found in high-sulfide sediments ($50\text{--}2,000 \mu M$), and *N. diversicolor* showed a broader distribution relative to pore-water sulfide.

DISCUSSION

Sulfide is naturally present in the top layers of marine and freshwater sediments, where most benthic organisms live. It is toxic to a variety of benthic animals. Sulfide forms complexes and insoluble metal sulfides with a variety of metals and hence affects their toxicity. Under sulfide stress, benthic tube dwellers may enhance irrigation with overlying water and hence reduce their exposure to other contaminants in sediments and pore waters.

However, we have at best only a limited understanding of such interactions involving sulfide in sediments. This presents two major problems related to determining the toxicity of sediments and the causative agents of such toxicity. First, we do

not know how important sulfide toxicity is to resident populations; second, by not adequately considering sulfide toxicity, we risk misidentifying the causative agents. It was only relatively recently that ammonia was recognized as an important sediment toxicant [48]. We now have procedures to control its toxicity in laboratory tests [49] and to assess its relative contribution to sediment toxicity compared to, for example, metals or synthetic organic contaminants [8]. In a draft U.S. Environmental Protection Agency report, Ankley et al. [50] recognized the possible contribution of sulfide and proposed several sediment toxicity identification evaluation manipulations. However, these procedures have not been effectively applied. It should be noted that differentiating sulfide toxicity is more difficult than differentiating ammonia toxicity, because any manipulations attempting to remove sulfide will also change metal speciation and toxicity.

We are fooling ourselves if we believe we are adequately assessing sediment toxicity in laboratory tests and appropriately assigning probable causative factors in surveys of field populations if we do not adequately account for possible sulfide toxicity. Sulfide is clearly a significant potential toxicant in marine sediments. Even in freshwater sediments, where sulfide concentrations are generally not elevated, the potential contribution of sulfide to toxicity cannot be neglected because freshwater animals are more sensitive than marine animals to sulfide.

Current standardized procedures for conducting sediment toxicity tests [51,52] cannot effectively be used to evaluate or identify the toxicity of sulfide. Although efforts can be taken during sediment sampling, transportation, and storage (e.g., introducing a nitrogen atmosphere) to minimize the loss of sulfide via volatilization and oxidation, current whole-sediment toxicity test procedures require aeration (or renewed oxidized water) and generally last at least 10 d [51,52], which will result in oxidation of most of the sulfide originally present in the sediments in addition to changing metal speciation. This problem is typically exacerbated when pore-water toxicity testing is used, because sulfide in water is more readily oxidized (at 25°C and pH 8.0, the half-life for sulfide oxidation is 50 h in air-saturated water and 26 h in air-saturated seawater [18]). The pore water being tested is hence not the same as in nature because of the loss of sulfide and changed speciation of metals. Although testing under a nitrogen atmosphere can maintain sulfide concentration and metal speciation, oxygen deficiency will kill most test organisms. Other changes in pore-water chemistry and toxicity have been observed elsewhere [53].

But even if the objective of sediment toxicity testing is only to test the toxicity of contaminants other than sulfide (which is presently the case whether researchers recognize it or not), there is still a problem. Specifically, aeration will not remove all sulfide in the sediment or pore water, and the remaining sulfide may still be sufficient to cause toxicity, particularly in short-term, sensitive tests [20].

Two options to identify sulfide toxicity in sediments are available. One option is to measure sulfide concentrations in situ and compare them with sulfide toxicity thresholds for the organisms of concern, either using literature data or generating such data using procedures such as those developed by Vismann [23]. Concentrations of either dissolved sulfide in pore water or AVS in sediments can be reasonably measured; however, toxicity thresholds for most benthic animals are unknown. The other option may be to assess the in situ abundance of sulfide-sensitive species. Lack of sulfide-sensitive species

might indicate toxicity related to sulfide, although it is often difficult to accurately ascribe faunal distributions to a single contaminant. However, again, data on sulfide tolerances are scanty.

Warren et al. [45] have shown that, even if sediment is heavily contaminated in situ, tube dwellers can thrive because, by bioirrigation, they actually live in microenvironments similar to the overlying water. We need to know whether, as a result of sulfide (and low-oxygen) stress, the common reaction for most benthic tube dwellers will be to increase the intensity of bioirrigation. If so, then bioirrigation is a presently underestimated factor in assessing the biological effects of sediment toxicants to resident populations. This response is ignored in laboratory toxicity and bioaccumulation tests, in which the overlying water is more similar to the surrounding sediments and pore water than would be the case in nature.

Neglecting sulfide as a toxicant of concern (e.g., not conducting an appropriate toxicity identification evaluation) can result in the cause of toxicity being misidentified (e.g., incorrectly ascribed to other contaminants). Such misidentification of a potential problem and its potential cause can have serious ramifications. For instance, sediments may inappropriately be identified as toxic because of persistent contaminants, such as metals, when in fact the toxicity is due to sulfide. Dredging of sediments contaminated with persistent compounds, such as metals or synthetic organics, can be problematic at best. In contrast, sulfide-contaminated sediments can be rapidly rendered harmless by mixing with well-oxygenated waters. Remedial approaches may thus be inappropriate and resources and effort expended on problems that are relatively easy to solve, or real problems may not be properly identified.

Clearly, there is an imperative need, for very pragmatic reasons, to develop appropriate tools and procedures to properly assess the toxicity of sulfide in aquatic sediments. We provide both short- and long-term recommendations for reaching this goal.

RECOMMENDATIONS

Aside from the obvious generic recommendation that sulfide in sediments needs to be specifically studied relative to both in situ faunal distributions and sediment toxicity testing, we make the following specific recommendations, which are intended not only to assist in the short term with survey and test interpretation but also to provide a database for long-term assessment of sulfide in sediments.

First, faunal surveys and sediment collections (i.e., for toxicity testing) should measure sulfide concentrations in pore waters as soon as the sediments are collected.

Second, toxicity identification evaluation procedures need to be further developed to adequately assess the possibility of sulfide toxicity, not just on disturbed sediments but also on undisturbed sediments.

Third, sulfide concentrations should be monitored in both overlying and pore water during whole-sediment toxicity testing and in pore water during pore-water toxicity testing when there is reason to believe this substance may be present at toxic levels (e.g., when the samples smell of sulfide). In such cases, this should be done, at the very least, before the start and at the end of testing.

Fourth, toxicity thresholds for sulfide need to be determined for a wide range of sediment-dwelling organisms, including both plants and animals, and particularly for organisms living in areas with relatively high sulfide levels.

Fifth, more laboratory and, above all, field research is needed to understand and quantify the contribution to metal toxicity of AVS in sediment and of sulfide in pore waters.

Sixth, the potential influence of sulfide stress on the intensity of bioirrigation needs to be more fully understood to determine which organisms are actually affected by sediment as opposed to overlying water sulfide (and other contaminant) concentrations.

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